1. Angle Quest: Deep Learning Approach for Tooth Segmentation and Axial Inclination Measurement in Orthopantomogram Radiographs

Methodology: Fully automated deep learning + computer vision framework for instance-level tooth segmentation and mesiodistal axial inclination angle calculation.

Data Collection: 110 OPG radiographs, Age group: 18–30 years, Ethical clearance obtained, Sample size statistically validated.

Annotation Strategy: LabelMe-based polygon annotation, Manual split-mask refinement, Converted into COCO format for Mask R-CNN training.

Data Augmentation: Horizontal flip, Vertical flip, Rotation (±15°), Brightness/contrast adjustment, Gaussian blur. Final dataset: 110 original + 220 augmented = 330 images.

Segmentation Model Selection: U-Net failed in overlapping roots and low contrast regions. Final Model: Mask R-CNN (ResNet-50 + FPN, Detectron2, mask_rcnn_R_50_FPN_3x). Training: 1000 iterations, Batch size 2, 128 RoI proposals, T4 GPU.

Loss Function: L(total) = L(RPN-cls) + L(RPN-box) + L(ROI-cls) + L(ROI-box) + L(mask).

Post-Processing: Binary conversion, Morphological operations, Otsu thresholding, CCA, Noise filtering (area < 2000 pixels removed).

Axis Construction: Minimum-area rectangle, top & bottom midpoints computed, vertical tooth axis generated.

Reference Lines: Orbitale-to-orbitale (maxillary), Mental foramen-to-foramen (mandibular).

Statistical Validation: Shapiro–Wilk test, Levene’s test, Unpaired t-test ($p < 0.05$).

Metric	Value
Precision	0.83
Recall	0.92
Dice Coefficient	0.87
IoU	0.77

2. Multiclass Segmentation using Teeth Attention Modules for Dental X-ray Images

Architecture: U-Net-like encoder-decoder, Swin Transformer, Teeth Attention Block (TAB), Multiscale Supervision, Squared Dice Loss.

Preprocessing: Resize to 1024x512, Normalize [0,255], Multiscale morphology operations, Noise reduction and contrast enhancement.

Dataset: 540 panoramic dental X-ray images, 32 tooth classes (FDI notation #11–#18, #21–#28, #31–#38, #41–#48). Split: 70% training, 15% validation, 15% testing.

Metric	Proposed Model
Accuracy (ACC)	0.9726
Dice Similarity Coefficient (DSC)	0.9102
Jaccard Index (JI)	0.8501
Precision	0.8046
Recall	0.9389
Specificity	0.9730

Variation	DSC
Basic U-Net	0.7602
+ Deep Supervision	0.7846
+ Swin Transformer	0.7644
+ TAB	0.9001
Full Proposed Model	0.9102

3. Boundary Feature Fusion Network for Tooth Image Segmentation

BFFNet: ResNet Backbone, Boundary Feature Extraction Module (Reverse Attention), Feature Cross-Fusion Module, Weighted IOU + Weighted BCE Loss, Deep supervision.

Dataset: STS (MICCAI 2023 Challenge), 2000 panoramic dental images.

Metric	BFFNet
Dice	0.7911
IOU	0.9848
Hausdorff Distance	0.0174
Overall Score	0.9061

4 & 5. Integrated Segmentation and Detection Models for Dentex Challenge 2023

Authors: Lanshan He, Yusheng Liu, Lisheng Wang

Affiliation: Shanghai Jiao Tong University

1. Objective

Integrated segmentation + detection framework to detect abnormal teeth, predict disease labels, and assign enumeration IDs (FDI notation) from panoramic dental X-rays. Developed for Dentex Challenge 2023.

2. Overall Methodology (Pipeline)

Step 1: Tooth Detection

Step 2: Disease Detection

Step 3: Label Matching

Separate detection heads used for tooth and disease detection, merged later for flexibility and simplified model design.

3. Dataset & Preprocessing

Subset 1: Quadrant labels

Subset 2: Tooth labels (converted to global 1–32 enumeration)

Subset 3: Disease labels (only disease ID retained for training)

Annotations converted to COCO and YOLO format.

Segmentation preprocessing: masks from JSON, resized to 256x256, augmentations (flip, crop, rotation).

Detection preprocessing: default model settings.

4. Tooth Detection

Detection Model: DINO (ResNet-50 backbone, COCO pretrained).

Segmentation Models: U-Net, SE U-Net.

Strategy 1: Whole image segmentation (32 classes).

Strategy 2: Quadrant-wise segmentation using DiffusionDet for quadrant detection, 9-class per quadrant training.

Segmentation masks converted to bounding boxes via largest connected component extraction.

5. Disease Detection

Model 1: DINO (Swin Transformer backbone), transfer learning from tooth detection.

Model 2: YOLOv8 trained directly on disease dataset.

Ensembling using Weighted Boxes Fusion (WBF).

6. Label Matching

IOU-based weighted voting assigns enumeration IDs.

Detection outputs weighted higher than segmentation outputs.

7. Post-Processing

Enumeration converted to FDI notation.

Duplicate removal, highest confidence retention.

Prior dental knowledge filtering (e.g., impacted teeth only 8th tooth).

8. Disease Detection Performance

Model	AR	AP	AP50	AP75
DINO	0.544	0.343	0.508	0.411
YOLOv8	0.530	0.352	0.539	0.399
Fused	0.592	0.371	0.543	0.447

9. Final Challenge Test Set Results

Label Type	AR	AP	AP50	AP75
Quadrant	0.753	0.474	0.679	0.585
Enumeration	0.681	0.353	0.511	0.421
Disease	0.592	0.371	0.543	0.447

Key Contributions:

Hybrid segmentation + detection framework.

Quadrant-wise segmentation for improved localization.

DINO + YOLOv8 fusion improves performance.

IOU-based voting for enumeration.

Dental prior knowledge in postprocessing.

Limitations:

Enumeration AP relatively low (0.353).

Disease AP modest (0.371).

Complex multi-stage pipeline.

Segmentation accuracy not independently evaluated.

6. A Deep Learning Approach to Teeth Segmentation and Orientation from Panoramic X-Rays

Authors: Mou Deb, Madhab Deb, Mrinal Kanti Dhar

1. Objective

Simultaneous teeth instance segmentation and orientation estimation using a two-stage framework (Segmentation + PCA-based OBB).

2. Dataset

DNS Dataset: 543 images, resolution 1991x1127, 5-fold split.

Patch size: 512x512, overlap 10x10, stride 502.

32 FDI labels, normalized to [0,1].

3. Model Architecture

Modified FUSegNet with EfficientNet-B7 encoder (ImageNet pretrained).

Grid-based Attention Gates.

Parallel Spatial & Channel Squeeze-and-Excitation (P-scSE).

~1.5% IoU improvement over original FUSegNet.

4. Training Details

Optimizer: Adam, LR=0.001, weight decay=1e-5.

50 epochs with LR scheduling.

Loss: Dice Loss + Focal Loss.

5. Post-Processing

Three merging cases for multiple regions per label.

Improved IoU from 82.0 to 82.43 and DSC from 90.1 to 90.37.

6. PCA-Based Oriented Bounding Box (OBB)

Extract tooth mask → PCA → compute principal component → derive angle → rotate → compute HBB → rotate back.

No ground-truth orientation required.

More interpretable and efficient than minAreaRect.

7. Segmentation Results

Model	IoU	DSC
DeepLabV3+	80.45	89.17
FPN	81.75	89.76
FUSegNet	80.79	89.38
UNet	67.93	80.90
MiT-b2-UNet	72.65	84.16
Swin-Unet	53.69	69.87
Proposed	82.43	90.37

8. OBB Results

Metric	Value
RIoU	82.82%
False Positives	0
False Negatives	3–6

9. Model Statistics

Parameters: ~66M

Inference time: 1.2 seconds per image

10. Challenges Identified

Tooth overlaps, foreign objects, fuzzy roots, poor annotations, low-quality images.

Manual review identified 21 foreign-object cases, 19 overlaps, 10 fuzzy roots.

Final Achievements:

IoU: 82.43%

DSC: 90.37%

RIoU: 82.82%

Zero false positives in OBB.

First integrated segmentation + orientation framework with PCA-based OBB.

7. Self-Supervised Learning with Masked Image Modeling for Teeth Numbering, Detection of Dental Restorations, and Instance Segmentation in Dental Panoramic Radiographs

Authors: Amani Almalki, Longin Jan Latecki

1. Objective of the Paper

This paper applies self-supervised learning (SSL) using Masked Image Modeling (MIM) to improve teeth numbering (FDI system), teeth detection, dental restorations detection, and instance segmentation.

Key motivation: Very limited dental radiograph datasets. Transformers such as Swin require large-scale training data. MIM is used to pre-train on the same dental dataset.

This is the first work applying SimMIM and UM-MAE to Swin Transformer on dental panoramic X-rays.

2. Main Contributions

Applied SimMIM and UM-MAE to Swin Transformer.

Demonstrated SimMIM significantly improves AP.

Corrected and augmented DNS dataset.

Added new annotations for dental restorations: Direct restorations, Indirect restorations, Root canal therapy.

Created TNDRS dataset (Teeth Numbering, Detection of Restorations, and Segmentation).

3. Dataset

Base Dataset: DNS dataset (543 panoramic X-rays). Resolution: 1991 × 1127. FDI numbering labels.

Limitations: No instance overlapping, systematic annotation errors, no dental restoration segmentation.

Dataset Enhancement (TNDRS): Fixed instance overlapping, corrected segmentation errors under dentist supervision, added restoration segmentation.

4. Methodology

Two-stage framework: Stage 1: Self-Supervised Pre-training using Swin Transformer encoder with SimMIM and UM-MAE.

Stage 2: Downstream tasks using Swin-B backbone, Cascade Mask R-CNN framework, FPN neck.

Tasks: Detection, Instance segmentation, Teeth numbering.

5. Self-Supervised Methods

SimMIM: Keeps masked tokens, preserves spatial structure, mask ratio 20%, patch size 16×16, L1 loss, 100 epochs.

UM-MAE: Drops masked tokens, reorganizes visible patches, mask ratio 25%, MSE loss, 800 epochs.

SimMIM maintains patch location which is critical in dental images.

6. Experimental Setup

5-fold cross-validation. Test: 111 images. Train: 432 images.

Image resized to 800×600. Batch size: 432.

Optimizer: AdamW. Fine-tuning LR: 0.0001. Weight decay: 0.05.

Augmentation: Noise addition, Horizontal flipping with renumbering.

Evaluation: APbox and APmask.

7. Results Before Restoration Augmentation

Initialization	APbox	APmask
PANet (CNN)	75.4	73.9
Random Swin	75.7	74.8
Supervised Swin	79.1	78.3
UM-MAE	84.5	83.2
SimMIM	86.1	84.6

SimMIM improves +10.4 APbox over random and +7.0 APbox over supervised initialization.

8. Results After Adding Dental Restorations

Initialization	APbox	APmask
Random	77.0	76.1
Supervised	80.3	79.2
UM-MAE	88.3	85.7
SimMIM	90.4	88.9

Final best results: 90.4% APbox and 88.9% APmask.

9. Mask Ratio Effect

Mask Ratio	APbox	APmask
60%	84.3	83.2
50%	84.7	83.6
40%	85.5	83.9
30%	85.9	84.1
20%	86.1	84.6
10%	85.8	84.3

Lower mask ratio works better because dental features are small and high masking removes too much detail.

10. Dataset Correction Impact

Condition	APbox	APmask
Before correction	80.2	78.2
After correction	86.1	84.6

Improvement: +5.9 APbox and +6.4 APmask due to annotation correction.

11. Efficiency Comparison

Method	Time	Memory
SimMIM	24.6h	18.7GB
UM-MAE	12.5h	6.7GB

UM-MAE is 2× faster and uses 2× less memory, but SimMIM performs better.

12. Limitations

Small dataset (543 images). SSL trained on same dataset. Computationally expensive. No orientation estimation.

8. S-R2F2U-Net: A Single-Stage Model for Teeth Segmentation

Authors: Mrinal Kanti Dhar, Mou Deb

Objective: Achieve high Dice, reduce parameters, improve over R2U-Net and state-of-the-art models.

Dataset: 1500 panoramic X-rays. Train: 911, Val: 153, Test: 436. Patch size 512×512 with 10×10 overlap.

Hybrid Loss: $L = \lambda 1 * \text{CrossEntropy} + \lambda 2 * \text{Dice}$. Best $\lambda 1=1$, $\lambda 2=0.5$.

Main Results

Model	Dice	Params (M)
Attention U-Net	92.55	43.94
U2-Net	91.78	60.11
R2U-Net	93.11	108.61
ResUNet-a	92.66	4.71
TSASNet	92.72	78.27
S-R2F2U-Net	93.26	59.12

Final Metrics: Accuracy 97.31%, Specificity 98.55%, Precision 94.27%, Recall 92.61%, Dice 93.26%.

9. DE-KAN: Dual Encoder Kolmogorov Arnold Network for Accurate 2D Teeth Segmentation

CDPR Dataset Results

Model	mIoU	Dice	Accuracy	Recall
U-KAN	86.50	91.79	95.95	92.20
Teeth U-Net	87.96	92.21	96.09	92.78
YOLOv8	85.91	90.90	93.30	91.68
DE-KAN	94.5	97.10	98.91	97.36

HTL Dataset Results

Model	mIoU	Dice
U-KAN	83.79	89.30
Teeth U-Net	84.27	89.69
YOLOv8	82.14	89.47
DE-KAN	89.45	94.39

Computational Analysis

Model	GFLOPs	Params (M)	Latency (ms)
U-KAN	16.44	25.36	37.82
YOLOv8	26.20	36.90	44.27
Teeth U-Net	75.24	36.04	55.63
DE-KAN	164.42	32.73	59.05

10. Tooth Instance Segmentation on Panoramic Dental Radiographs Using U-Nets and Morphological Processing

Authors: Selahattin Serdar Hell■, Andaç Hamamc■ (2022)

Dataset: 105 panoramic X-rays resized to 512×512. Two mask types: Full mask and Split mask.

Experiments (10-fold CV)

Experiment	Mask Type	Augmentation	Post-Processing
E1	Full	No	No
E2	Split	No	No
E3	Split	H+V+Noise	No
E4	Split	H+Noise	No
E5	Split	No	Yes

Dice Results

Exp	Dice (%)
E1	94.5 ± 0.7
E2	95.1 ± 0.5
E3	95.2 ± 0.3
E4	95.4 ± 0.3
E5	95.3 ± 0.6

Tooth Counting Error Reduced from 26.81% to 6.15% with morphological processing (p=0.002).